

REGULATION OF SERCA BY MEMBRANE POLYPEPTIDES

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Keywords: SERCA, Calcium transport, regulins, NMR spectroscopy

ABSTRACT

Maximum 1 page. (font : cmr10, size : 12)

Calcium homeostasis is fundamental to heart muscle contraction and is primarily maintained by the sarcoplasmic reticulum Ca^{2+} -ATPase (SERCA), which is responsible for the bulk of intracellular calcium transport in humans. The activity of SERCA is finely regulated by a family of single-pass membrane peptides known as regulins, which ensure calcium levels remain within a precise physiological range. Disruption of this regulation has been implicated in the development of cardiomyopathies and heart failure. Among these regulins, phospholamban and DWORF play crucial but opposing roles in the beat-to-beat modulation of cardiac contractility. Unphosphorylated phospholamban inhibits SERCA by decreasing its affinity for calcium, whereas phosphorylation by protein kinase A lifts this inhibition and restores SERCA function. In contrast, DWORF enhances SERCA activity independently of β -adrenergic signaling. Notably, overexpression of DWORF can displace phospholamban and rescue impaired cardiac contractility, yet the molecular mechanism underlying SERCA activation by DWORF remains poorly understood. Here, we employ NMR spectroscopy and molecular dynamics simulations to dissect the molecular mechanisms governing both the activation and deactivation of cardiac SERCA. Our findings offer new mechanistic insights that could inform the development of targeted therapies for cardiomyopathies stemming from aberrant calcium homeostasis.

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